

① Find the first palindromic string in array.

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Sol:- `def first_palindrome(words):`

`for word in words:`

`if word == word[::-1]:`

`return word`

`return ""`

`word = ["abc", "car", "ada", "racecar"]`

`print("First palindromic string:", first_palindrome(words))`

Output:-

First-palindromic string: ada

② selection sort with Sample Array

Code:-

`arr = [64, 25, 12, 22, 11]`

`for i in range(len(arr)):`

`min_idx = i`

`for j in range(i+1, len(arr)):`

`if arr[j] < arr[min_idx]:`

`min_idx = j`

`arr[i], arr[min_idx] = arr[min_idx], arr[i]`

`print("Sorted Array:", arr)`

Output:-

Sorted Array: [11, 12, 22, 25, 64]

⑤ Merge Sort with Comparison Count Code

Code:-

```
count = 0
```

```
def merge_sort(arr):
```

```
    global count
```

```
    if len(arr) > 1:
```

```
        mid = len(arr) // 2
```

```
        left = arr[:mid]
```

```
        right = arr[mid:]
```

```
        merge_sort(left)
```

```
        merge_sort(right)
```

```
        i = j = k = 0
```

```
        while i < len(left) and j < len(right):
```

```
            count += 1
```

```
            if left[i] < right[j]:
```

```
                arr[k] = left[i]
```

```
                i += 1
```

```
            else:
```

```
                arr[k] = right[j]
```

```
                j += 1
```

```
                k += 1
```

```
        while i < len(left):
```

```
            arr[k] = left[i]
```

```
arr = [5, 2, 4, 1, 3]
```

```
print("sorted array:", arr)
```



```
print ("comparisons:", count)
```

Output:-

Sorted Array: [1, 2, 3, 4, 5]

Comparisons: 8

④ Dice Throw problem Using Dynamic Programming

$n = 2$

target = 7

$dp = [0] * (target + 1)$

$dp[0] = 1$

for i in range(n):

 new ~~to~~ [0] * (target + 1)

 for s in range(target + 1)

 for face in range(1, 7)

 if $s + face \leq target$

$new[s + face] += dp[s]$

$dp = new$

print ("Number of ways", $dp[target]$)

Output:-

Number of ways: 6

Possible ways

Code:

(1, 6)

(2, 5)

(3, 4)

(4, 3)

(5, 2)

(6, 1)